

Program 10:

Demonstrate minikube dashboard using CLI and Kubernetes Cluster

Project Structure:

program_10/

—>deployment.yaml

—>services.yaml

Procedure:

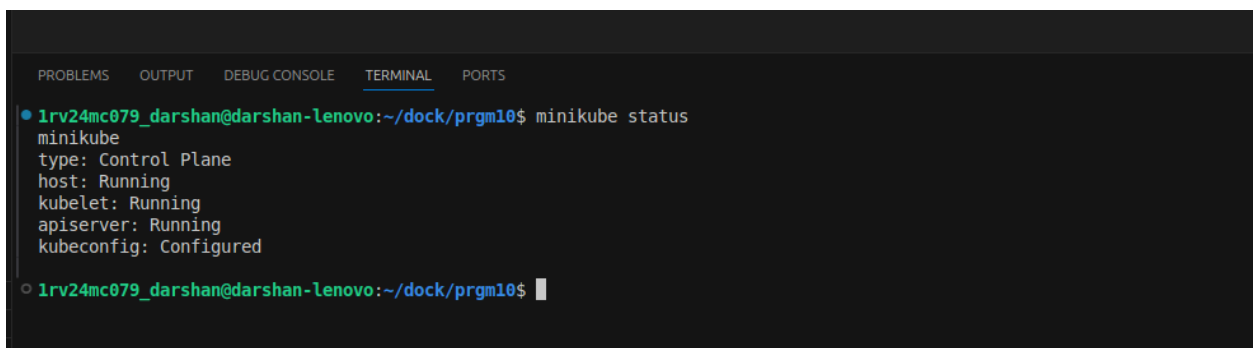
Create a folder (ex: program_10)

Step 1: Start Minikube

minikube start

Check the status of Minikube

minikube status

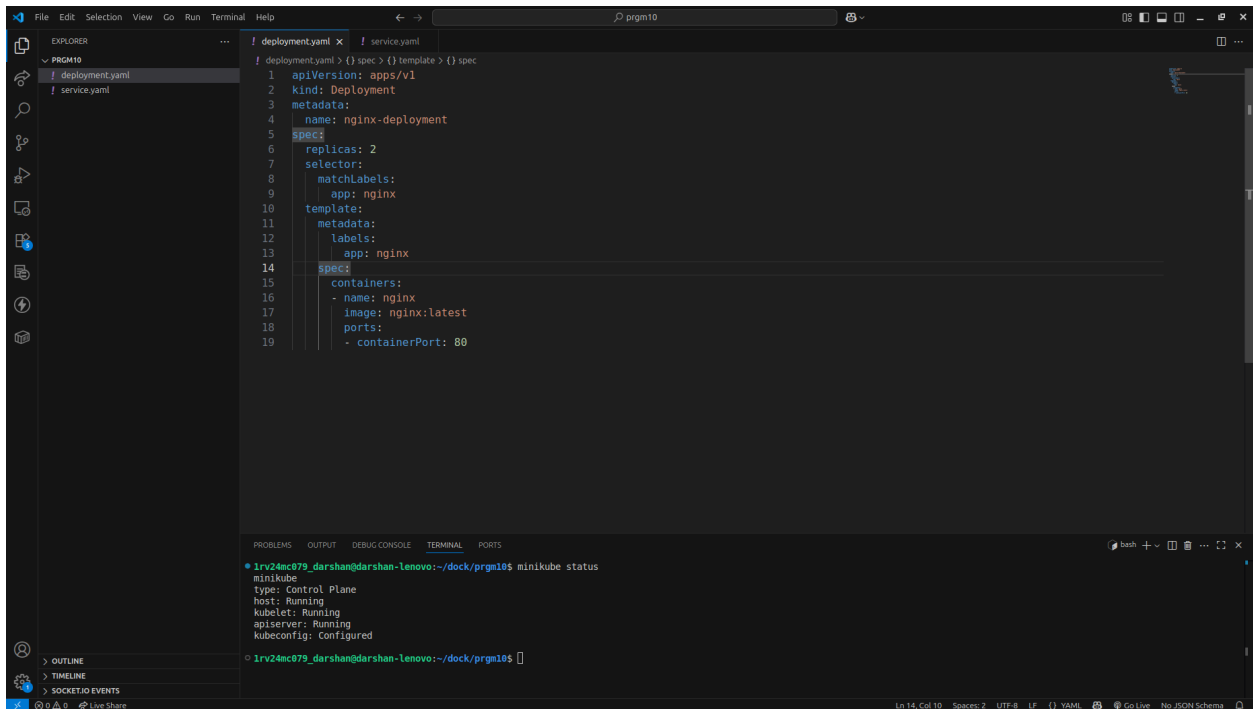
A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active and underlined), and 'PORTS'. The terminal shows a command prompt where the user has entered 'minikube status'. The output of the command is displayed as follows: 'minikube', 'type: Control Plane', 'host: Running', 'kubelet: Running', 'apiserver: Running', and 'kubeconfig: Configured'. Below the output, there is another prompt line showing the user's shell prompt and the current directory.

```

1rv24mc079_darshan@darshan-lenovo:~/dock/prgm10$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
1rv24mc079_darshan@darshan-lenovo:~/dock/prgm10$

```

Step 2: Create deployment.yaml file with the following code
nano deployment.yaml



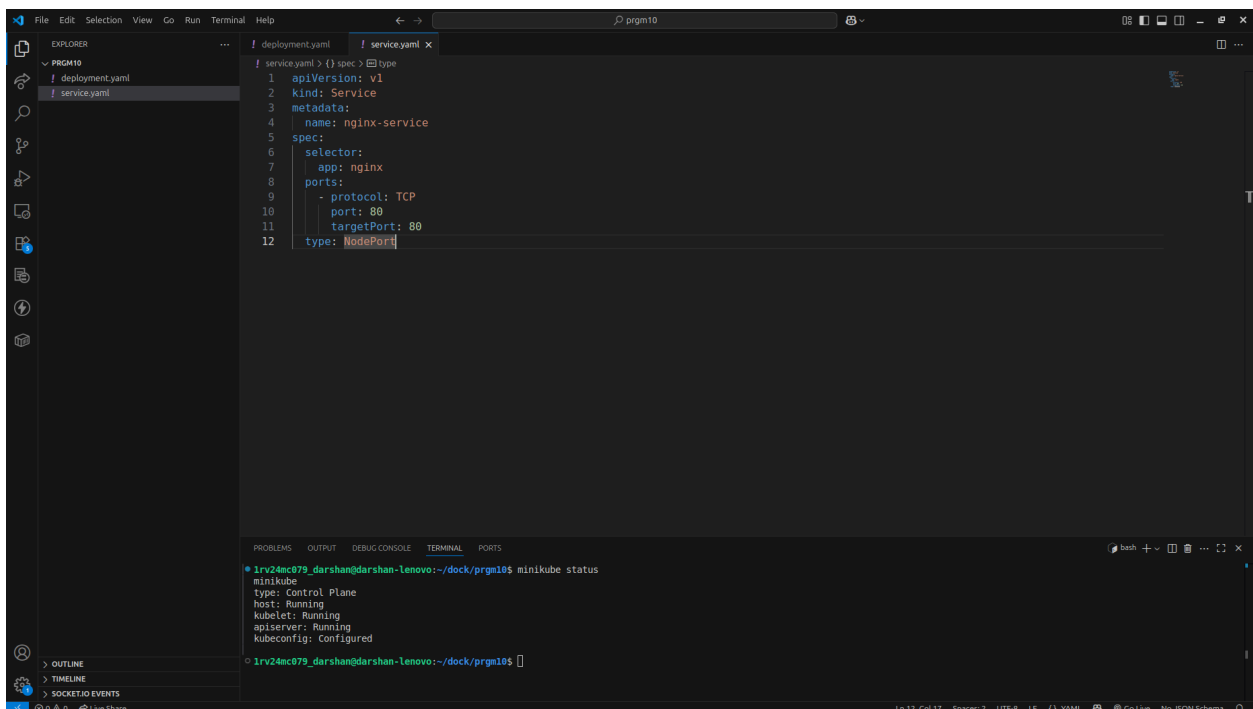
The screenshot shows the VS Code editor with the `deployment.yaml` file open. The file content is as follows:

```
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: nginx-deployment
5 spec:
6   replicas: 2
7   selector:
8     matchLabels:
9       app: nginx
10  template:
11    metadata:
12      labels:
13        app: nginx
14  spec:
15    containers:
16      - name: nginx
17        image: nginx:latest
18        ports:
19          - containerPort: 80
```

The terminal at the bottom shows the command `minikube status` being executed, with the following output:

```
minikube
Type: Control Plane
Host: Running
Kubelet: Running
APIServer: Running
Kubeconfig: Configured
```

Step 3: Create services.yaml file with the following code
nano services.yaml



The screenshot shows the VS Code editor with the `services.yaml` file open. The file content is as follows:

```
1 apiVersion: v1
2 kind: Service
3 metadata:
4   name: nginx-service
5 spec:
6   selector:
7     app: nginx
8   ports:
9     - protocol: TCP
10     port: 80
11     targetPort: 80
12 type: NodePort
```

The terminal at the bottom shows the command `minikube status` being executed, with the following output:

```
minikube
Type: Control Plane
Host: Running
Kubelet: Running
APIServer: Running
Kubeconfig: Configured
```

Step 4: Apply the files

kubectl apply -f deployment.yaml

kubectl apply -f services.yaml

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
• 1rv24mc079_darshan@darshan-lenovo:~/dock/prgm10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment configured
• 1rv24mc079_darshan@darshan-lenovo:~/dock/prgm10$ kubectl apply -f service.yaml
service/nginx-service configured
○ 1rv24mc079_darshan@darshan-lenovo:~/dock/prgm10$
```

Step 5: Get pods, deployment, services

kubectl get pods

kubectl get deployment

kubectl get services

Step 6: Check the minikube services

minikube services nginx-service

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

service

Run 'minikube --help' for usage.
• 1rv24mc079_darshan@darshan-lenovo:~/dock/prgm10$ minikube service nginx-service


| NAMESPACE | NAME          | TARGET PORT | URL                       |
|-----------|---------------|-------------|---------------------------|
| default   | nginx-service | 80          | http://192.168.49.2:30969 |


🚀 Opening service default/nginx-service in default browser...
○ 1rv24mc079_darshan@darshan-lenovo:~/dock/prgm10$ Opening in existing browser session.
```